COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 3 : SLO – 1 | Bus structures and its types
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Section A – Match the Following (Answers)

• 1. Data Bus → B. Transfers data between components
• 2. Address Bus → C. Transfers address information
• 3. Control Bus → A. Transfers control signals
• 4. CPU → D. Executes instructions and manages data flow
• 5. Single Bus Structure → E. One shared line for all devices

Section B – Fill in the Blanks

• A BUS (single bus) connects CPU, memory, and I/O devices through a common communication line.
• In a bus system, the ADDRESS BUS is responsible for transferring address information.
• The CONTROL bus is used to carry control signals such as read or write commands.
• A SINGLE bus system allows only one device to use the bus at any given time.
• One of the disadvantages of a single bus structure is BUS CONTENTION (slower communication due to sharing), which can degrade performance.

Section C – Activity Task

• Single Bus Structure (described): CPU, Memory and I/O devices all connected to one shared bus, which logically consists of the Data Bus, Address Bus and Control Bus lines.
• Why only one device at a time: all units share the same set of lines; if two devices drove the bus simultaneously their signals would clash, so an arbiter/the CPU grants the bus to one transfer at a time.
• Effect on performance: transfers are serialized — only one data movement can occur per bus cycle, creating a bottleneck; however the design is simple and low-cost.

Section D – True or False

• The data bus transfers control signals. — FALSE
• A single bus structure is more cost-effective than a multiple bus structure. — TRUE
• In a single bus system, multiple devices can simultaneously communicate using the bus. — FALSE
• The CPU manages and controls the flow of data in a bus system. — TRUE
• A fault in the bus line can cause the entire system to fail. — TRUE